

CLINICAL INVESTIGATION

Eye

VISUAL ACUITY AFTER RUTHENIUM¹⁰⁶ BRACHYTHERAPY OF CHOROIDAL MELANOMASBERTIL DAMATO, PH.D., F.R.C.OPHTH.,* IMRAN PATEL, M.SC.,[†] IAN R. CAMPBELL, B.SC.,[‡]
HELEN M. MAYLES, M.SC.,[†] AND R. DOUGLAS ERRINGTON, F.R.C.R.[§]*Ocular Oncology Service, St. Paul's Eye Unit, Royal Liverpool University Hospital, Liverpool, UK; [†]Physics Department and[§]Radiotherapy Department, Clatterbridge Centre for Oncology, Bebington, Wirral, UK; and [‡]IC Statistical Services, Wirral, UK**Purpose:** To report on conservation of visual acuity after Ruthenium¹⁰⁶ (Ru-106) brachytherapy of choroidal melanoma.**Methods and Materials:** This study was a noncomparative interventional case series of 458 patients with choroidal melanoma treated at a single center between January 1993 and December 2001. The intervention consisted of Ru-106 brachytherapy delivering minimum scleral and apex doses of 300 Gy and 80 Gy, respectively, using a 15-mm or 20-mm plaque. For discrete, posterior tumors, the plaque was positioned eccentrically with its posterior edge aligned with the posterior tumor margin. To ensure correct plaque positioning, any overlying extraocular muscles were dis-inserted, and the locations of both tumor and plaque edges were confirmed by transillumination and indentation. The main outcome measures were conservation of vision of 20/40 or better, 20/200 or better, and Counting Fingers or better, according to baseline variables.**Results:** The actuarial rate of conservation of 20/40 or better was 55% at 9 years, loss of such vision correlating with posterior tumor extension ($p < 0.001$), temporal tumor location ($p = 0.001$), increased tumor height ($p = 0.01$), and older age ($p < 0.01$) (Cox multivariate analysis). Similar analyses showed conservation of 20/200 or better in 57% of eyes at 9 years, loss correlating with reduced initial visual acuity ($p < 0.001$), posterior tumor extension ($p < 0.001$), and temporal tumor location ($p = 0.006$). Counting Fingers or better vision was conserved in 83% of patients at 9 years, loss correlating with increased tumor height ($p < 0.0001$). Local tumor recurrence occurred in 9 patients (actuarial rate, 3% at 9 years).**Conclusion:** Ruthenium¹⁰⁶ brachytherapy of posterior choroidal melanoma achieves good conservation of vision if the tumor does not extend close to the optic nerve or fovea. © 2005 Elsevier Inc.

Ruthenium brachytherapy, Choroidal melanoma, Visual acuity.

INTRODUCTION

Conservative treatment of choroidal melanoma is aimed at preserving as much useful vision as possible while eradicating any threat of metastatic spread. In most centers, the first choice of treatment is brachytherapy using either an iodine-125 or Ruthenium¹⁰⁶ (Ru-106) plaque. Visual outcome depends on a variety of factors, which include tumor height, proximity to optic disk and fovea, tumor shape, patient age at treatment, and systemic diseases such as diabetes mellitus (1–14).

To avoid marginal tumor recurrence, the conventional practice is to center the plaque over the tumor base so that it overlaps all the visible tumor margins by at least 2 mm (10, 12); however, this causes visual loss if the tumor extends close to the optic disk or fovea. Since 1993, when treating choroidal melanomas extending post-equatorially,

we have whenever possible placed the plaque eccentrically in relation to the tumor so that it was as far as possible from the optic nerve and fovea, thereby reducing chances of visual loss. In a previous study, we have shown that our rates of local tumor recurrence and enucleation compare favorably with those reported in other studies (15).

The main aim of this study was to document visual outcome in patients with choroidal melanoma treated with Ru-106 brachytherapy at our center. A further aim was to identify risk factors for visual loss.

METHODS AND MATERIALS

Patients and data collection

Patients were included in the study prospectively and consecutively if they were treated for choroidal melanoma between 1993, when the Liverpool Ocular Oncology Centre was established, and

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